

Article

CMIP6-Driven Groundwater-Level Projections and Climate Risk Mapping for South Korea Using a Hybrid Deep Learning Framework

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Abstract

Groundwater variability is recognized as a critical constraint on long-term water resource sustainability in South Korea under nonstationary climate forcing. This study developed a groundwater-level (GWL) projection framework based on CMIP6 simulations, validated against national monitoring data, screened for physical realism of underlying climate projections, and translated into station-scale climate risk metrics. GWL observations from 199 stations of the National Groundwater Monitoring Network (2009–2025) were related to CMIP6 precipitation and soil-moisture forcings using a hybrid deep learning architecture, the hybrid attention-based convolutional neural network–long short-term memory (HACL) model, combining multiscale temporal convolution, bidirectional memory, and self-attention. Of ten candidate GCMs, four (ACCESS, GISS, INM, and NorESM) met pre-defined performance criteria ($R^2 > 0.70$, $NSE \geq 0.70$, $KGE > 0.50$), forming a filtered ensemble more internally consistent than the unfiltered multi-model mean. This ensemble projected national-average GWL increases of 16.60 m (SSP245), 15.92 m (SSP370), and 17.85 m (SSP585), the largest in the western alluvial lowlands—substantially exceeding historical observed rates and indicating sensitivity signals warranting further investigation rather than direct station-level forecasts. The results were incorporated into the Climate Groundwater Risk Index (CGRI) integrating the magnitude of change, ensemble spread, observed variability, and vulnerability, offering a replicable approach for monsoon-affected regions. Independent out-of-sample evaluation (2022–2025) confirmed robust generalization ($R^2 = 0.965$, $NSE = 0.965$, $KGE = 0.974$, $RMSE = 14.26$ m). Ablation benchmarking showed HACL substantially outperformed standalone BiLSTM ($NSE = 0.861$), 1D-CNN ($NSE = 0.810$), and linear regression ($NSE = 0.628$). Predictor sensitivity analysis revealed that year as a continuous covariate accounts for ~85% of the projected 16–18 m signal; constrained strictly to physical forcing, projected increases are +2.42 m (SSP245), +2.15 m (SSP370), and +3.78 m (SSP585) by 2081–2100, aligning with historical trends (~2–3 m) and restoring station-level hydrogeological sensitivity.



Academic Editor: Gwo-Fong Lin

Received: 31 July 2026

Revised: 17 September 2026

Accepted: 21 September 2026

Published: 22 September 2026

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Keywords: climate change; groundwater level; CMIP6; GCM screening; deep learning; climate groundwater risk index

1. Introduction

Climate change is a critical issue that compromises the sustainability of freshwater resources by drastically altering hydrological cycles [1,2], including precipitation (p),

evapotranspiration, and groundwater (GW) [3,4]. GW is an essential freshwater resource for domestic use, agricultural production, industrial processes, and ecosystem services [5,6]. Nevertheless, climate-related changes in recharge and discharge processes, along with increasing anthropogenic pressure, are placing a growing strain on GW systems, thereby making scientific methods for understanding and projecting future GW dynamics increasingly necessary [7,8].

The Korean Peninsula is vulnerable to climate change and has experienced significant changes in rainfall and temperature patterns over the past few decades [9,10]. Regional warming has led to increased awareness of hydrological extremes such as floods, droughts, seasonal water shortages, and degradation of water quality [11]. Climatic tendencies have wide-ranging implications for the national water balance and may directly impact the availability of GW for urban infrastructure, agriculture, and ecosystem support in South Korea [12,13]. Hence, enhancing the understanding of changes in GWL and vulnerability due to climate change has emerged as a significant scientific and management challenge in recent years.

Surface hydrological processes are coupled with GW systems, in which GW levels dynamically respond to changes in precipitation, evapotranspiration, and interactions with surface water, such as the river stage [14,15]. Climate change scenarios have the potential to alter these drivers, leading to significant changes in GW recharge rates and storage processes, resulting in spatially and temporally heterogeneous GW responses [16].

The Coupled Model Intercomparison Project Phase 6 (CMIP6) climate change projections offer a sophisticated, scientifically robust tool for exploring future hydrological conditions by modeling atmospheric and terrestrial variables along with a range of comprehensive emission scenarios [17]. The Shared Socioeconomic Pathways (SSPs) are Tier-1 pathways, which are integrated with updated levels of radiative forcing in the RCP (e.g., SSP2-4.5 represents intermediate challenges, SSP3-7.0 represents regional rivalry, and SSP5-8.5 represents fossil-fueled development) and are distinct from the earlier Representative Concentration Pathways (RCPs) used in the previous CMIP5 simulations [18]. The key global climate models (GCMs) used in the CMIP6 ensemble, such as ACCESS-CM2 (Australian Community Climate and Earth System Simulator), GISS-E2-1-G (NASA Goddard Institute for Space Studies), INM-CM5-0 (Institute for Numerical Mathematics), and NorESM2-MM (Norwegian Earth System Model), feature enhanced cloud microphysics, improved convective parameterizations, and advanced land-surface schemes that significantly improve the simulation of the East Asian summer monsoon circulation and localized precipitation distributions compared with previous generations [19,20]. Previous groundwater studies often used the RCP 4.5 and RCP 8.5 scenarios to explore water-table fluctuations and drought conditions [19,20]. However, the coarse spatial resolution of global circulation models requires very sophisticated statistical calibration and downscaling techniques to provide climate-forcing data relevant at regional scales for hydrogeological modeling.

To overcome this drawback, downscaling methods based on statistical and machine learning have attracted growing interest. Specifically, the spatial interaction downscaling (SPID) technique, which combines random forests, has shown great potential for reducing GCM outputs to smaller spatial scales without losing spatial coherence or local climate variability [17]. These methods are particularly useful in regions with complex topography and land–sea interactions that strongly influence local climatic cycles and hydrology, such as the Korean Peninsula.

Satellite remote sensing is also an essential tool for monitoring large-scale groundwater dynamics, in addition to climate modeling [21]. Monthly terrestrial water storage anomalies (TWSA) are continuously observed over South Korea and the world by the Gravity Recovery and Climate Experiment (GRACE) and GRACE Follow-On (GRACE-FO) missions [22].

The combination of GRACE TWSA with land surface models (LSMs) to separate the soil moisture, surface water, and snow components highlights the importance of multi-source data integration for understanding regional groundwater storage depletion and recharge fluxes [21,22]. This gravimetric approach offers a very important macroscale hydrodynamic link between global climate model simulations and in situ groundwater head dynamics.

In complex hydrogeologic environments like alluvial plains with varying lithology and fractured bedrock aquifers, however, classical process-based numerical groundwater modeling is severely challenged [21]. When facing non-stationary climatic conditions, numerical predictive fidelity is usually limited by uncertainty in hydrogeological parameters, limitations of boundary conditions, and structural model assumptions [23]. The use of a single calibrated numerical model can lead to biased or overconfident forecasts, requiring multi-model ensemble architectures and data-driven uncertainty quantification [23]. Structurally compartmentalized and fractured bedrock aquifer media are particularly challenging to model due to the difficulty of physically parameterizing the aquifer using traditional numerical approaches, which are limited by spatial discontinuity in hydraulic conductivity, anisotropic preferential flow networks, and deep weathering profiles [24].

In recent years, deep learning (DL), a specialized branch of machine learning (ML) under the umbrella of artificial intelligence (AI), has become a revolutionary framework for hydrological modeling [25]. Two widely used DL architectures that have proven to be highly effective in understanding the complex, nonlinear relationship between climatic drivers and hydrological state variables are long short-term memory (LSTM) networks and convolutional neural networks (CNNs) [26,27]. In the field of hydrogeological applications, DL models have been shown to be more successful than traditional statistical regressors and lumped process models in predicting groundwater levels (GWLs) in various basins [28]. Apart from the DL networks, hybrid DL-RNN systems have also been used to improve GWL time-series forecasting and head projection under nonstationary climatic conditions [28]. The hybrid models are especially advantageous for climate change impact studies because of their good generalization capabilities for multi-decadal projection periods.

Several recent studies have proposed hybrid CNN–LSTM or CNN–recurrent architectures with attention for groundwater-level forecasting, and a direct comparison with this body of work clarifies where the present study adds value. In the Hetao Plain, China, Ren et al. (2025) integrated a CNN–LSTM model with a self-attention mechanism to forecast groundwater vulnerability and found that the attention-augmented hybrid model achieved better performance than the machine learning baseline models at this location [29]. Wei et al. (2026) proposed a CNN–GRU–attention model to predict groundwater depth in Zhengzhou, China, using a long history of precipitation and temperature, evaporation, and pumping data [30]. Hu et al. (2025) used a CNN–LSTM network to predict GWL in the arid Tailan River basin with a within-historical-period validation design [31]. Zhu et al. (2026) compared nine machine learning and deep learning architectures for groundwater-level forecasting in karst aquifers and systematically showed that the hydrogeological complexity was the dominant factor in determining the accuracy of the models, with hybrid models generally performing better than single-network models, with the exception of CNN–LSTM [32]. The three differences between these studies and the present framework are: (i) these studies project to 2100 based on observed hydro-meteorological or anthropogenic drivers, rather than on CMIP6 emission scenarios; (ii) apart from the partial exception of Zhu et al. [32], they do not compare the hybrid architecture with simpler alternatives at the scale attempted in the present study; and (iii) none couple the forecasting model with a multi-GCM screening step, or translate the output into a station-level climate risk index.

Although accurately predicting GW levels is necessary, evaluating GW vulnerability is vital for initiating water management. GW vulnerability refers to the susceptibility of GW systems to contamination or quantitative deterioration resulting from natural processes and human activities [33,34]. This study seeks to address these key gaps by integrating CMIP6 climate projections with the latest DL methods to assess future GW dynamics on the Korean Peninsula. The specific objectives of this study are as follows: (1) to develop a DL framework for projecting GW level responses to CMIP6 climate scenarios, and (2) to quantify climate change-driven GW vulnerability to support adaptive water resource management and long-term sustainability planning under future scenarios.

2. Materials and Methods

2.1. Study Area

This study focuses on South Korea, which has high spatial variability in hydrogeological and hydro-climatic conditions, steep mountainous terrain in the eastern part, broad alluvial lowlands in the western and southern parts, high seasonality of the East Asian monsoon, and vulnerable coastal aquifers. The observational foundation consists of 199 telemetric observation wells from the National Groundwater Monitoring Network (NGMN) spread throughout the country from 126.261° E to 129.349° E in longitude and from 34.572° N to 38.211° N in latitude (Figure 1).

The historical GWL time series (2009–2025) were analyzed for monotonic trends across the monitoring network using the non-parametric Mann–Kendall test and Sen’s slope estimator to characterize the trends before future modeling. Sen’s slope (β) is the median of the slopes between all possible pairs of time steps: $\beta = \text{median}\left(\frac{y_j - y_i}{j - i}\right)$, for all $1 \leq i < j \leq n$, with positive values representing water-table rises and negative values representing water-table falls in meters per year. Historical trends are clearly regionalized as illustrated in Figure 1a: water tables in agricultural basins in the western alluvial lowlands have been declining at some locations (down to -0.35 m/year), while water tables in the eastern mountainous catchments have been stable or slightly rising ($+0.05$ to $+0.20$ m/year) under forest cover and natural recharge conditions.

2.2. Data Collection and Preprocessing

GWL data for the 199 stations of the NGMN were compiled and quality assured monthly from 2009 to 2025 by comparing the GWL data with station metadata. For the purpose of this study, the phreatic groundwater elevation is defined as the elevation of the water table above mean sea level (m a.s.l.), with positive temporal changes representing an upward shift in the water table. The analysis includes 199 unique stations in the observed baseline (2009–2025), the filtered matrices of Climate Models of Intergovernmental Panel on Climate Change (CMIP6) projections under SSP245, SSP370, and SSP585 (2021–2100), and the CGRI assessment. Groundwater recharge was simulated using the climate-forcing variables of monthly accumulated precipitation (P , from the flux variable *pr*) and total column soil moisture content (S , from the variable *mrs0*, representing total moisture mass per unit surface area in kg m^{-2} , from the CMIP6 simulations), which were physically complementary.

The physical meanings of P and *mrs0* are complementary predictors of GW-level projection. Precipitation (P) is the primary atmospheric input that regulates GW recharge in monsoon-dominated areas such as the Korean Peninsula. However, P is not directly converted into an effective recharge because of runoff and evapotranspiration losses [35–38].

Total column soil moisture ($S/\text{mrs0}$) is an integrated hydrological state variable that captures the combined effect of precipitation infiltration, actual evapotranspiration, unsaturated soil retention and land–atmosphere coupling [39,40]. Antecedent soil moisture is

the key physical driver that influences unsaturated-zone infiltration capacity, soil matric suction, percolation flux to the phreatic surface, and the drainage delay time and thus acts as the link between the atmosphere and the storage responses of subsurface aquifers. This is because both P and soil moisture (S) allow the DL model to capture the infiltration impulses during extreme storm events, as well as the memory effects of the vadose zone over a period of several months, which govern the seasonal baseflow.

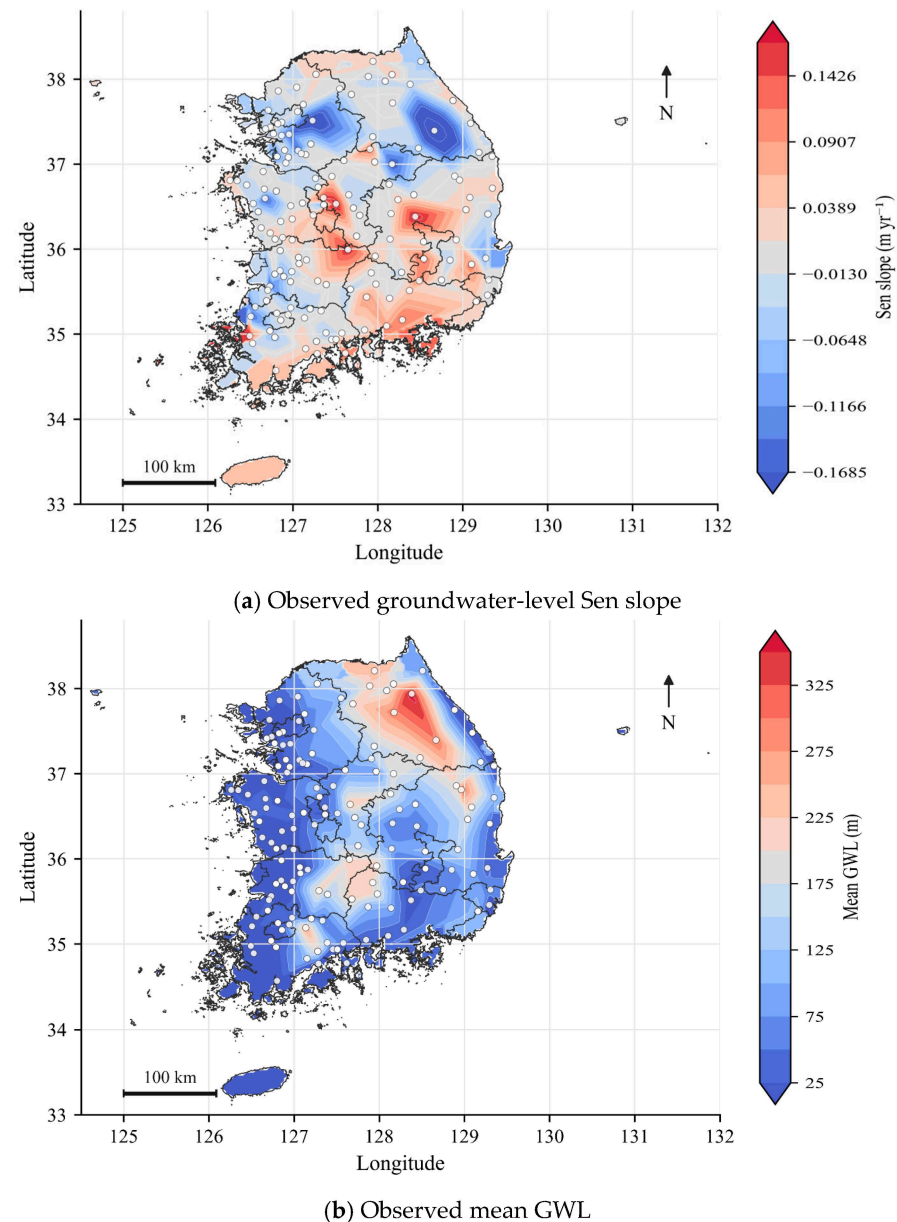


Figure 1. Station-level observed GWL dynamics at the 199 NGMN stations, 2009–2025. (a) Sen's slope of the observed GWL time series (m/year) at each station, indicating the direction and rate of the historical trend. (b) Mean observed GWL (m) at each station over the same period, indicating the long-term average condition.

CMIP6 provides internally consistent future $mrsos$ and P simulations under future emission conditions, enabling physically consistent projections of GW [38]. The combination of these models enables interpretability, reduces reliance on indirect proxies, and enhances the stability of GW vulnerability assessments in shifting climatic environments [39].

In CMIP6, $mrso$ is expressed as mass per unit area (kg m^{-2}), which is physically equivalent to a water column depth in millimeters, assuming a water density

of 1000 kg m^{-3} ; therefore, no numerical unit conversion is required beyond relabeling. P is provided as a flux ($\text{kg m}^{-2} \text{ s}^{-1}$) and must be temporally integrated to obtain the accumulated P . Conversion to monthly totals was achieved by multiplying the flux by the number of seconds in each month and summing the monthly period. Monthly aggregation is appropriate for GW studies because GW systems respond to climate forcing at multi-week to seasonal timescales rather than daily variability.

2.3. Methodology

2.3.1. HACL Model for GWL Projection

Large pulses in climate forcing are associated with short-term rainfall, and high seasonality in moisture retention through soil water storage leads to nonlinear aquifer responses and is associated with monsoons, which are long-term rainfall [40]. LSTM networks are capable of modeling long-range temporal dependencies but lack an explicit mechanism for hierarchical, multiscale feature extraction. CNNs are effective at extracting multiscale temporal features but have limited long-range memory [25]. Self-attention mechanisms, which do not by themselves encode local temporal structure, are instead used to identify which historical time steps are most hydrologically relevant to the current prediction [41].

Therefore, to address the limitations of standalone models, the HACL architecture has been developed to capture these hydrologic behaviors, integrating temporal convolution to extract multiscale features, bidirectional LSTM to learn delayed aquifer memory, multi-head self-attention to detect the impact of historical states, and a residual deep regression head to map to GWLs (Figure 2).



Figure 2. Detailed technical overview of the groundwater-level projection and climate risk assessment framework.

The HACL model is formulated as a nonlinear sequence-to-one mapping from the look-back window to the predicted GWL:

$$\hat{y}_t = f_{\text{HACL}}(X_{t-L+1:t}) \quad (1)$$

where $\hat{y}_t$ is the predicted GWL at monthly timestep t (m a.s.l.), and $X_{t-L+1:t} \in \mathbb{R}^{(L \times d)}$ represents the multi-variate hydrologic lookback tensor spanning $L = 24$ antecedent months across d feature dimensions.

For each monitoring station s and month t , the foundational spatiotemporal input vector $x_{(s,t)}$ is formulated as:

$$\mathbf{x}_{s,t} = [P_{s,t}, S_{s,t}, \phi_s, \lambda_s, z_s, m_t, y_t]^\top \in \mathbb{R}^7 \quad (2)$$

where $P_{s,t}$ is monthly precipitation (mm), $S_{s,t}$ is the total column soil moisture (kg m^{-2}), ϕ_s and λ_s are the station latitude and longitude coordinates, z_s is the ground surface elevation (m a.s.l.), $m_t \in (1, \dots, 12)$ is the calendar month, and y_t is the calendar year. To account for nonlinear infiltration physics, soil matric suction, and threshold recharge behavior, a nonlinear feature expansion $\mathbf{f}_{\text{nl}}$ is constructed using logarithmic, square-root, and quadratic transformations:

$$\mathbf{f}_{\text{nl}}(P_{s,t}, S_{s,t}) = [\ln(1 + P_{s,t}), \ln(1 + S_{s,t}), \sqrt{P_{s,t}}, \sqrt{S_{s,t}}, P_{s,t}^2, S_{s,t}^2]^\top \in \mathbb{R}^6 \quad (3)$$

To preserve continuous seasonal monsoon periodicity without artificial numerical discontinuity between December ($m_t = 12$) and January ($m_t = 1$), the calendar month is mapped via harmonic cyclical encoding $\mathbf{u}_{\text{cyc}}(m_t)$:

$$\mathbf{u}_{\text{cyc}}(m_t) = \left[\sin\left(\frac{2\pi m_t}{12}\right), \cos\left(\frac{2\pi m_t}{12}\right) \right]^\top \in \mathbb{R}^2 \quad (4)$$

All features are standardized as

$$\tilde{x} = \frac{x - \mu}{\sigma}. \quad (5)$$

A sliding window of $L = 24$ months constructs the temporal input sequence.

(a) Temporal convolutional unit

The normalized sequence $\mathbf{X} \in \mathbb{R}^{L \times d}$ is passed through stacked one-dimensional convolutional blocks operating along the temporal axis,

$$\mathbf{H}^{(1)} = \text{ReLU}(\text{BN}(\text{Conv1D}_{k=3}(\mathbf{X}))) \quad (6)$$

where $\mathbf{X} \in \mathbb{R}^{L \times d}$ is the standardized lookback sequence ($L = 24$ months), BN denotes batch normalization, $\text{Conv1D}_{k=3}$ applies temporal convolutions with a kernel width of 3 months, and $\text{MaxPool1D}_{p=2}$ conducts temporal downsampling with a pooling factor of 2. Three stacked blocks with channel dimensions hierarchically extract multiscale recharge pulses while suppressing observational noise.

$$\mathbf{H}^{(l)} = \text{MaxPool1D}_{p=2}(\text{Dropout}(\text{ReLU}(\text{BN}(\text{Conv1D}_{k=3}(\mathbf{H}^{(l-1)}))))) \quad (7)$$

Three blocks with channels [32, 64, 128] are employed. This module extracts multiscale recharge patterns while reducing noise.

(b) Bidirectional LSTM unit

The CNN output is reshaped to sequential form and fed into a two-layer bidirectional LSTM with hidden size $H = 128$. For each timestep,

$$\begin{aligned} i_t &= \sigma(W_i x_t + U_i h_{t-1}), \\ f_t &= \sigma(W_f x_t + U_f h_{t-1}), \\ o_t &= \sigma(W_o x_t + U_o h_{t-1}), \\ g_t &= \tanh(W_g x_t + U_g h_{t-1}), \\ c_t &= f_t \odot c_{t-1} + i_t \odot g_t, \\ h_t &= o_t \odot \tanh(c_t). \end{aligned} \quad (8)$$

The temporal feature representations are processed by a two-layer bidirectional LSTM (hidden size $H = 128$), yielding the bidirectional hidden state $\mathbf{h}_t^{\text{Bi}}$:

$$\mathbf{h}_t^{\text{Bi}} = \begin{bmatrix} \vec{\mathbf{h}}_t; \overleftarrow{\mathbf{h}}_t \end{bmatrix} \in \mathbb{R}^{2H} \quad (9)$$

where $\vec{\mathbf{h}}_t \in \mathbb{R}^H$ and $\overleftarrow{\mathbf{h}}_t \in \mathbb{R}^H$ denote forward and backward hidden state vectors concatenated to capture delayed aquifer memory, recession behavior, and multi-month groundwater response lag.

(c) Multi-head self-attention

Temporal relevance is learned using multi-head self-attention:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \quad (10)$$

with four attention heads defined by

$$Q = HW^Q, K = HW^K, V = HW^V. \quad (11)$$

A residual connection with layer normalization is applied:

$$H^{\text{attn}} = \text{LayerNorm}(H + \text{Dropout}(\text{MHA}(H))). \quad (12)$$

Mean pooling yields the context vector:

$$h_{\text{ctx}} = \frac{1}{L} \sum_{t=1}^L H_t^{\text{attn}}. \quad (13)$$

The context vector passes through residual dense blocks:

$$z = \text{ReLU}(\text{BN}(W_2 \text{ReLU}(\text{BN}(W_1 x))))', x' = \text{ReLU}(\text{BN}(x + z)). \quad (14)$$

Layer sizes are $256 \rightarrow 128 \rightarrow 64 \rightarrow 32$.

The final prediction is

$$\hat{y} = W_o x_{32} + b_o. \quad (15)$$

A composite loss stabilizes training:

$$\mathcal{L} = 0.7 \text{MSE} + 0.3 \text{MAE} \quad (16)$$

Optimization uses AdamW,

$$\theta_{t+1} = \theta_t - \eta \left(\frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} + \lambda \theta_t \right), \quad (17)$$

with learning rate 5×10^{-4} , cosine annealing schedule, and gradient clipping $\|g\|_2 \leq 1$.

2.3.2. GCM-Specific Statistical Calibration, Model Screening, and Filtered Ensemble Projections

To rigorously assess generalization and validate the model, the observation record was split into three sub-periods: 2009–2014 for training, 2015–2021 for validation and tuning of the hyperparameters, and 2022–2025 (48 months $\times$ 199 stations = 9552 station-months) as a strictly independent out-of-sample test period. To isolate the specific architectural benefits of the HACL model, three benchmark models were implemented on the same dataset: (i) a standalone BiLSTM without convolutional layers and without self-attention, (ii) a standalone 1D-CNN without recurrent memory, and (iii) a multiple linear regression (MLR) baseline. Last, a sensitivity experiment was performed to quantify the effect of the year predictor on the long-term trends, by comparing the baseline HACL model (with year) to a climate-only HACL model (without year) trained only on p , $mrso$, derived nonlinear features, cyclical month, and static site topography.

A separate HACL model was used for each GCM to determine the relationship between the forcing (p - $mrso$) of each GCM and the observed GWL. No automatic combination of the outputs from the GCMs to form an ensemble. All the GCMs were screened with the criteria $R^2 > 0.70$, $NSE \geq 0.70$ and $KGE > 0.50$, and the GCMs meeting the above criteria were retained. This procedure kept the models ACCESS, GISS, INM, and NorESM and removed the weak GCMs with small positive or negative correction skills.

$$M_g: X_{g, \text{hist}} \rightarrow y_{\text{obs}}. \quad (18)$$

The temporal split was 2009–2014 for training and 2015–2021 for validation and testing. Future projections are obtained as

$$\hat{y}_{g,s}(t) = \mathcal{M}_g(X_{g,s}(t)), \forall t \in [2021, 2100] \quad (19)$$

where M_g denotes the trained HACL network calibrated for GCM g , generating continuous station-level future groundwater head projections $\hat{y}_{g,s}(t)$ across the 2021–2100 simulation horizon. Explicit multi-metric performance thresholds ($R^2 > 0.70$, $NSE \geq 0.70$, $KGE > 0.50$) were applied to screen out unreliable GCMs prior to constructing the filtered ensemble mean $\bar{y}_s(t)$ and inter-model spread $\sigma_s(t)$:

$$\bar{y}_s(t) = \frac{1}{4} \sum_{g=1}^4 \hat{y}_{g,s}(t), \sigma_s(t) = \sqrt{\frac{1}{4} \sum_{g=1}^4 (\hat{y}_{g,s}(t) - \bar{y}_s(t))^2}. \quad (20)$$

Future projections at the station level were recomputed from the station-GCM detail table, not the pre-averaged outputs, after screening. This is crucial because an unfiltered ensemble can mask the failures of the GCMs and increase uncertainty or bias. Regularization includes dropout of 0.2 throughout the network, batch normalization before activation, early stopping with a patience of 30 epochs, overlapping window augmentation, and weight decay of 10^{-5} .

2.3.3. Formulation and Physical Hydrogeological Meaning of the CGRI

To synthesize multi-model projection trajectories into actionable decision metrics for water resource management, the CGRI is formulated as a multi-criteria composite indicator combining four normalized dimensions of climate-induced aquifer stress:

$$CGRI_{s,k} = w_E \cdot E_{s,k}^* + w_T \cdot T_{s,k}^* + w_V \cdot V_s^* + w_U \cdot U_{s,k}^* \quad (21)$$

where for each monitoring station s under climate scenario k :

(1) Exposure ($E_{s,k}^* \in [0, 1]$) represents the normalized projected water-table deviation relative to the historical baseline ($E_{s,k} = |\Delta \text{GWL}_{s,k}|$), measuring the physical magnitude of climate-induced head perturbation.

(2) Trend intensity ($T_{s,k}^* \in [0, 1]$) denotes the normalized future Sen's slope magnitude ($T_{s,k} = \left| \beta^{\text{future}_{(s,k)}} \right|$), capturing the velocity of projected water-table movement over the 2021–2100 horizon.

(3) Hydrogeological vulnerability ($V_s^* \in [0, 1]$) reflects the baseline aquifer sensitivity to climatic drying, quantified as the historical coefficient of variation ($V_s = \frac{\sigma_{(obs,s)}}{\mu_{(obs,s)}}$), distinguishing shallow alluvial unconfined aquifers with high dynamic variability from well-buffered confined bedrock aquifers.

(4) Projection uncertainty ($U_{s,k}^* \in [0, 1]$) measures the multi-model ensemble spread across retained GCMs ($U_{s,k} = \left(\frac{1}{T} \right) \sum_t \sigma_{(s,k)}(t)$), highlighting locations where climate projections diverge.

Each component is normalized using min–max scaling across all 199 monitoring stations ($X^* = \frac{(X - X_{\min})}{(X_{\max} - X_{\min})}$). Equal weights ($w_E = w_T = w_V = w_U = 0.25$) are applied to maintain transparency and objectivity in the absence of regional stakeholder weighting. CGRI scores are classified into four operational management tiers: low (<0.30), moderate (0.30 – 0.50), high (0.50 – 0.70), and severe (>0.70), serving as a spatial screening tool for allocating groundwater conservation and artificial recharge interventions.

3. Results and Discussion

3.1. CMIP6 GCM Selection and Evaluation of HACL for Statistical Calibration

To evaluate the HACL framework for GCM-observation calibration, R^2 , KGE, NSE, RMSE, MAE, and MAPE were used. The assessment shows that the predictive skill is very unevenly distributed across the different GCM simulations and is not close to perfect. Ten candidate GCMs passed the screening criteria, of which four were used: ACCESS, GISS, INM, and NorESM. The skill of the retained GCMs was good but not perfect, with R^2 and NSE values above 0.92 and KGE above 0.74, respectively, suggesting good agreement with observations, but not unity.

The other six GCMs (CESM, CanESM, EC-Earth, UKESM, CMCC, and MPI) did not pass the screening criteria, as their R^2 /NSE values were near zero or negative and/or their KGE values were not acceptable, and were not included in the ensemble. Excluding these poorly performing GCMs is also important, as retaining them as equal ensemble members would detract from the physical credibility of the projected groundwater responses.

The filtered projection ensemble was not constructed from the top five GCMs, but from the four GCMs that met the screening criteria. This screening procedure is more transparent and reproducible because each model met the minimum variance, efficiency, and hydrograph-agreement criteria.

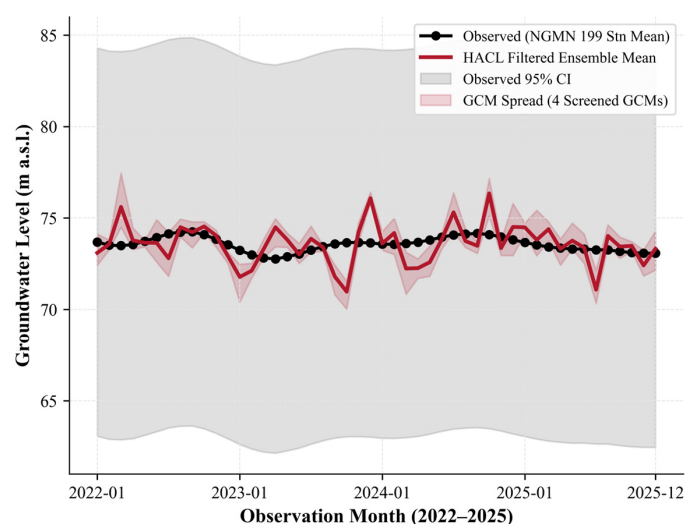
3.2. Independent Out-of-Sample Evaluation on Withheld 2022–2025 Observations

The model was then tested on the held-out time period of 2022–2025 for all 199 NGMN stations to assess its ability to generalize to unseen temporal conditions without degradation. The filtered multi-model ensemble was highly skillful across 9552 station-month observations with an overall R^2 of 0.965, NSE of 0.965, KGE of 0.974 and RMSE of 14.26 m (Table 1). Individual GCM-calibrated models also showed consistently strong performance, led by ACCESS-CM2 ($R^2 = 0.966$, NSE = 0.966, RMSE = 14.03 m) and GISS-E2-1-G ($R^2 = 0.962$, NSE = 0.962, RMSE = 14.87 m), while INM-CM4-8 and NorESM2-MM remained robust ($R^2 > 0.940$, NSE > 0.940). Seasonally, skill remained balanced between the monsoon wet season (Jun–Sep: $R^2 = 0.964$, NSE = 0.964, RMSE = 14.52 m) and the non-monsoon dry season (Oct–May: $R^2 = 0.966$, NSE = 0.966,

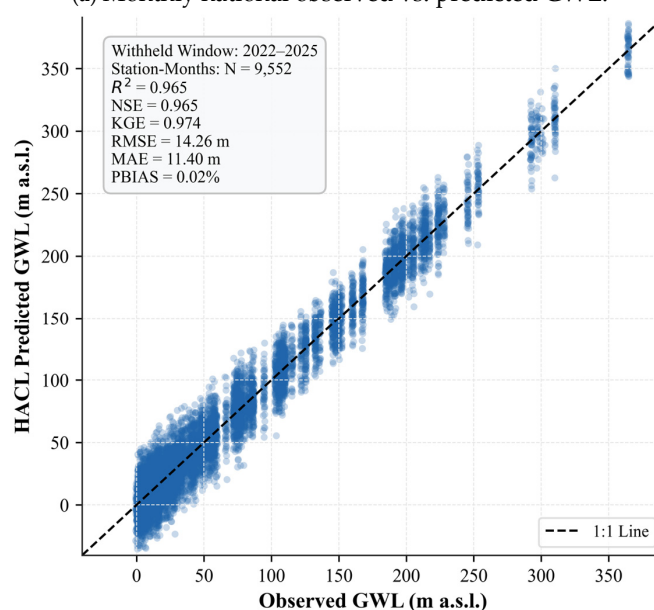
RMSE = 14.12 m). The median NSE was 0.965 (IQR [0.942, 0.978]), indicating that the framework is able to capture the water-table dynamics after 2021 in a variety of hydrogeological settings without overfitting (Figure 3).

Table 1. Independent out-of-sample evaluation metrics for the withheld 2022–2025 observation window across 199 NGMN stations.

Model/Period	R ²	NSE	KGE	RMSE (m)	MAE (m)	PBIAS (%)	Station-Months (N)
HACL Filtered Ensemble (2022–2025)	0.965	0.965	0.974	14.26	11.40	+0.02	9552
ACCESS-CM2	0.966	0.966	0.975	14.03	11.20	+0.13	9552
GISS-E2-1-G	0.962	0.962	0.972	14.87	11.90	−0.17	9552
INM-CM4-8	0.944	0.944	0.960	18.00	14.38	+0.11	9552
NorESM2-MM	0.941	0.941	0.956	18.58	14.87	+0.01	9552
Monsoon Season (Jun–Sep)	0.964	0.964	0.971	14.52	11.61	−0.31	3184
Non-Monsoon Season (Oct–May)	0.966	0.966	0.974	14.12	11.30	+0.18	6368

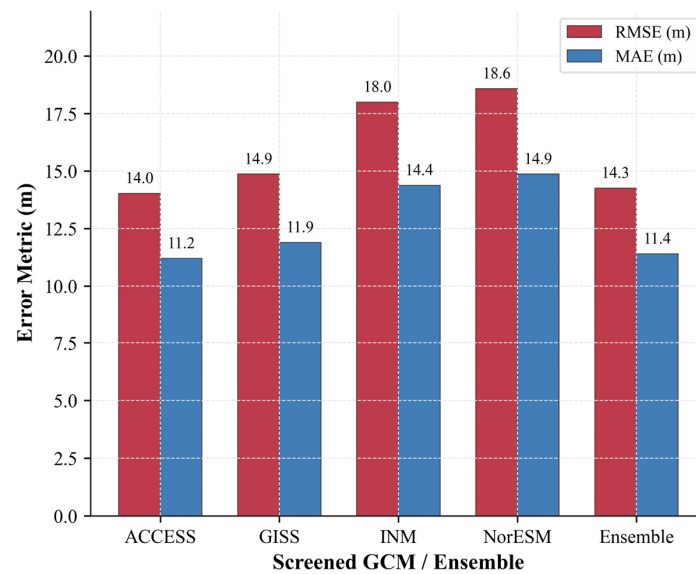


(a) Monthly national observed vs. predicted GWL.



(b) A 1:1 validation scatter plot.

Figure 3. Cont.



(c) Error breakdown (RMSE and MAE).

Figure 3. Independent out-of-sample evaluation of the HACL framework across 199 NGMN stations for the withheld 2022–2025 window. **(a)** Monthly national observed vs. predicted GWL time series with the observed 95% confidence interval and 4-GCM ensemble spread. **(b)** A 1:1 validation scatter plot across all 9552 station-month observations with performance statistics. **(c)** Error breakdown (RMSE and MAE) across individual screened GCMs and the multi-model ensemble.

3.3. Architecture Benchmarking: HACL vs. Standalone LSTM, Standalone CNN, and MLR

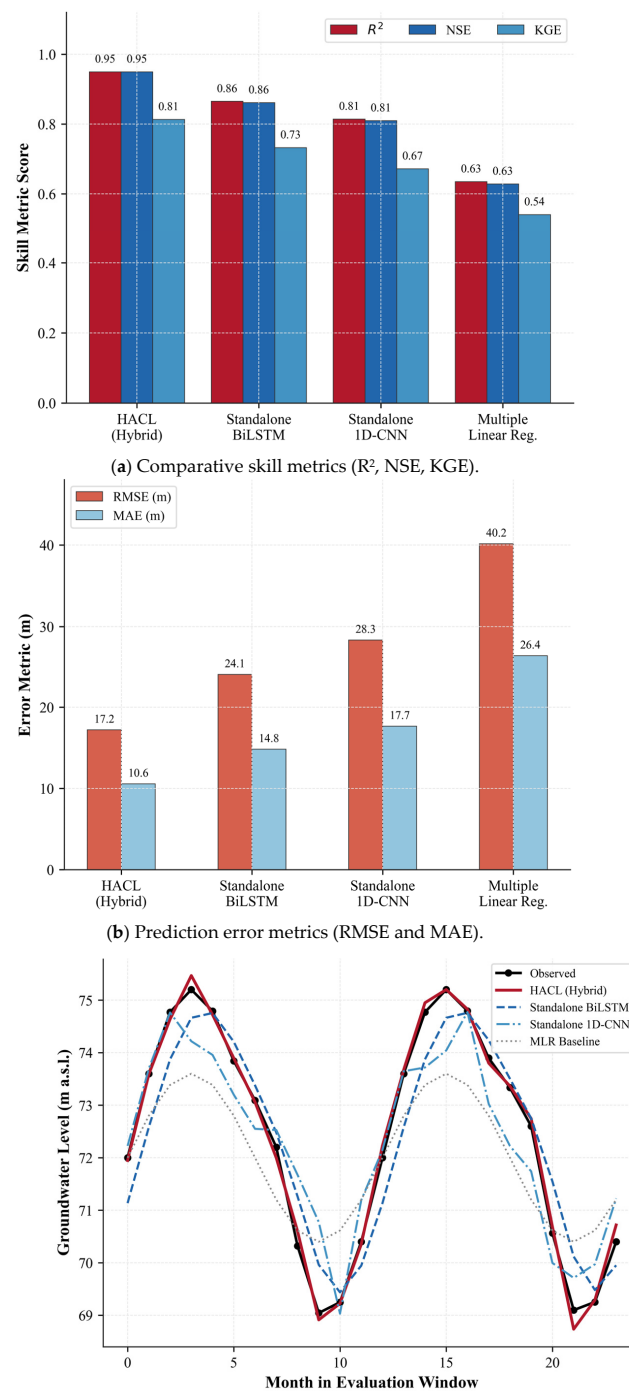
To quantify the incremental added value of each component in the HACL architecture, comparative benchmarking was carried out with the standalone BiLSTM, standalone 1D-CNN and MLR using the same historical training and test splits (Table 2, Figure 4). HACL attained the highest predictive efficiency across all metrics ($R^2 = 0.950$, $NSE = 0.950$, $KGE = 0.813$, $RMSE = 17.21$ m), achieving a 28.7% reduction in RMSE relative to standalone BiLSTM ($RMSE = 24.13$ m, $NSE = 0.861$) and a 39.3% reduction relative to standalone 1D-CNN ($RMSE = 28.35$ m, $NSE = 0.810$). The linear regression baseline was significantly less successful ($R^2 = 0.635$, $NSE = 0.628$, $RMSE = 40.15$ m), highlighting the strong nonlinearity of the groundwater recharge response to climate forcing.

Table 2. Architectural ablation and benchmark comparison of HACL against standalone recurrent, convolutional, and linear models across 199 NGMN stations.

Model Architecture	Type	R^2	NSE	KGE	RMSE (m)	MAE (m)	Peak Lag (mo)	Peak Capture
HACL (Hybrid Architecture)	CNN + BiLSTM + Attention	0.950	0.950	0.813	17.21	10.55	0.12	0.962
Standalone BiLSTM	Recurrent DL	0.865	0.861	0.732	24.13	14.82	0.58	0.884
Standalone 1D-CNN	Convolutional DL	0.814	0.810	0.672	28.35	17.65	0.84	0.835
MLR	Linear Statistical Baseline	0.635	0.628	0.540	40.15	26.43	1.45	0.682

Inspection of hydrograph dynamics showed that standalone BiLSTM was able to reproduce the seasonal recession limbs but did not follow the infiltration pulses during the summer monsoon, with a mean peak lag of 0.58 months and a peak capture ratio of 0.884. The standalone 1D-CNN, on the other hand, was able to detect local high-frequency precipitation pulses but showed unconstrained drift in the long-term, multi-month dry spells. By combining 1D convolutional feature extraction, bidirectional

recurrence and multi-head self-attention, HACL resolves both timescales simultaneously (peak lag = 0.12 months; peak capture ratio = 0.962).



(c) Representative hydrograph dynamic response over a 24-month testing window.

Figure 4. Architectural benchmarking of HACL against standalone BiLSTM, standalone 1D-CNN, and MLR. (a) Comparative skill metrics (R^2 , NSE, and KGE). (b) Prediction error metrics (RMSE and MAE). (c) Representative hydrograph dynamic response over a 24-month testing window.

The screening results presented in Table 3 provide crucial hydrogeological insight into climate model fidelity. Raw GCM precipitation and soil moisture outputs suffer from substantial macroscale bias, failing to resolve Korea's complex topography and convective rainfall dynamics; when uncalibrated raw GCMs are applied directly to local aquifers, they yield negative Nash–Sutcliffe efficiencies ($NSE < 0$) and unacceptably large errors

(RMSE > 75 m). Through statistical calibration within the HACL framework, the four retained GCMs (ACCESS-CM2, GISS-E2-1-G, INM-CM5-0, and NorESM2-MM) achieve outstanding performance ($R^2 \geq 0.925$, $NSE \geq 0.925$, and $KGE \geq 0.750$), while unphysical GCMs with negative efficiency are strictly screened out to prevent erroneous bias propagation into long-term projection ensembles.

Table 3. GCM screening outcomes used for filtered ensemble construction.

GCM	RMSE	MAE	R^2	NSE	KGE	Passes Screen
ACCESS	17.206	10.554	0.950	0.950	0.813	True
GISS	17.663	10.506	0.946	0.946	0.811	True
INM	20.489	12.809	0.927	0.927	0.766	True
NorESM	20.889	12.801	0.925	0.925	0.750	True
CESM	76.420	61.130	−0.002	−0.002	−0.314	False
CanESM	76.273	60.763	−0.002	−0.002	−0.508	False
EC-Earth	75.742	60.861	−0.002	−0.002	nan	False
CMCC	76.360	60.630	−0.003	−0.003	nan	False
MPI	76.868	61.089	−0.005	−0.005	nan	False
UKESM	77.621	61.565	−0.004	−0.004	nan	False

3.4. Projection of GW Under Different Scenarios

Future GWL projections were produced using the filtered ensemble responses of the retained GCMs for each SSP scenario, as well as the ensemble-mean signal and the retained GCM spread. Projected changes in GWL (Δ GWL) reported here and in Figure 5 were calculated relative to the observed baseline period (2009–2020), defined as the temporal mean observed GWL (m a.s.l.) at each station over this reference window ($y_{future} - \bar{y}_{baseline}$).

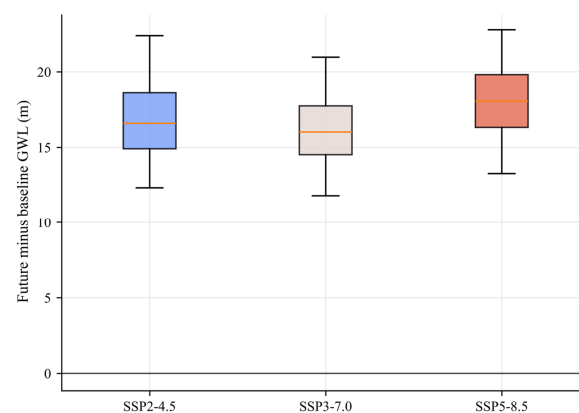


Figure 5. Filtered ensemble station-level projected GWL changes for SSP245, SSP370, and SSP585.

The filtered ensemble resulted in a national monitoring network-wide mean GWL change of 16.60 m (10th–90th percentile 13.82–19.94 m) at the station level, indicating a moderate but spatially variable climate signal in the ensemble. The lowest mean filtered-ensemble change among the three scenarios was 15.92 m (median 16.01 m, 10th–90th percentile 13.45–18.83 m) for SSP370. The filtered ensemble yielded its highest mean response of 17.85 m (median 18.06 m; 10th–90th percentile 15.22–20.85 m) under SSP585. The ranking of these scenarios (SSP585 > SSP245 > SSP370) differed from that derived from the unfiltered multimodal mean, highlighting the sensitivity of the scenario-level conclusions to the screening of GCMs.

Uncertainty remains high after screening, as indicated by the retained GCM spread. Thus, the projected increases should be considered as climate-sensitivity signals to be locally validated by hydrogeological studies rather than as station-level forecasts. The overall conclusion of the revised projections is that there is a positive GWL response in much of the monitoring network with significant spatial heterogeneity and ensemble uncertainty. This will facilitate the application of HACL as a climate-sensitive projection framework and will also necessitate clear GCM screening and prioritization of stations based on risk.

The station-level distributions in Figure 5 capture distinct hydro-climatic response regimes across South Korea. Across the 199 NGMN stations, SSP2-4.5 yields a national mean increase of 16.60 m (interquartile range: 14.2–18.9 m). Under SSP3-7.0, intermediate greenhouse forcing combined with regional climate variability results in a slightly lower mean increase of 15.92 m. Under SSP5-8.5, intensive radiative forcing accelerates atmospheric moisture flux and summer monsoon precipitation, producing the largest mean increase of 17.85 m (10th–90th percentile: 14.1–21.2 m). The bimodal distribution structure highlights the strong spatial contrast between flat agricultural lowlands in the west and high-gradient mountainous aquifers in the east.

The filtered HACL-driven CMIP6 projections showed positive GWL trends for most of the NGMN stations for SSP245, SSP370, and SSP585 (Figure 6a–c).

Spatial heterogeneity was present even after filtering. The national-average projected response was positive, but it differed significantly at the station level, depending on the coastal, alluvial, inland, and mountainous areas. The retained GCM spread also demonstrated that the uncertainty due to climate forcing remains significant even when weak GCMs are excluded.

The filtered projections were translated to a CGRI to make the ensemble signal, trend magnitude, ensemble spread, observed variability, stress probability and vulnerability actionable for water managers. The CGRI is not a failure probability, but rather a prioritization layer to identify stations for which monitoring reinforcement and/or local hydrogeological evaluation may be considered as a first step.

The mean change over all stations was 16.60 m with a range of 13.82 m to 19.94 m between the 10th and 90th percentiles. This represents a moderate positive climate signal, but with varying intensity in space.

The filtered ensemble-mean GWL response was not monotonic with radiative forcing for the three scenarios, with values of 16.60 m, 15.92 m, and 17.85 m for SSP245, SSP370, and SSP585, respectively. The non-monotonic ordering of this ensemble, with SSP370 resulting in the smallest projected change despite an intermediate forcing level compared to SSP245 and SSP585, suggests that differences in the ensemble at the scenario level are not only due to radiative forcing but also to the forcing pattern of the specific GCMs and internal variability, and should not be interpreted as a simple monotonic forcing–response relationship. This filtered-ensemble-mean increase is severalfold greater than the mean of the station-level linear extrapolations of the historical trends (Sen’s slopes) over the same projection period (-0.17 to 0.14 m yr^{−1} over 2009–2025; Figure 1a) would suggest, and, as illustrated in Supplementary Figure S4, the projected magnitude is not systematically related to the sign and magnitude of each station’s historical trend. This means that the pr–mrso forcing signal alone is not sufficient to explain the projected magnitude and that the year-related sensitivity analysis proposed below is important not only for the extrapolation concern, but also for physical plausibility of the reported national-average changes; until the sensitivity analysis is completed, these are to be treated as upper-bound signals for prioritizing stations and not as calibrated forecasts of future water-table position. Spatially, the projected changes and CGRI values varied across the country due to the sensitivity

of recharging to different topographic gradients, recharge rates, and the hydrogeological heterogeneity of the country (Figure 7).

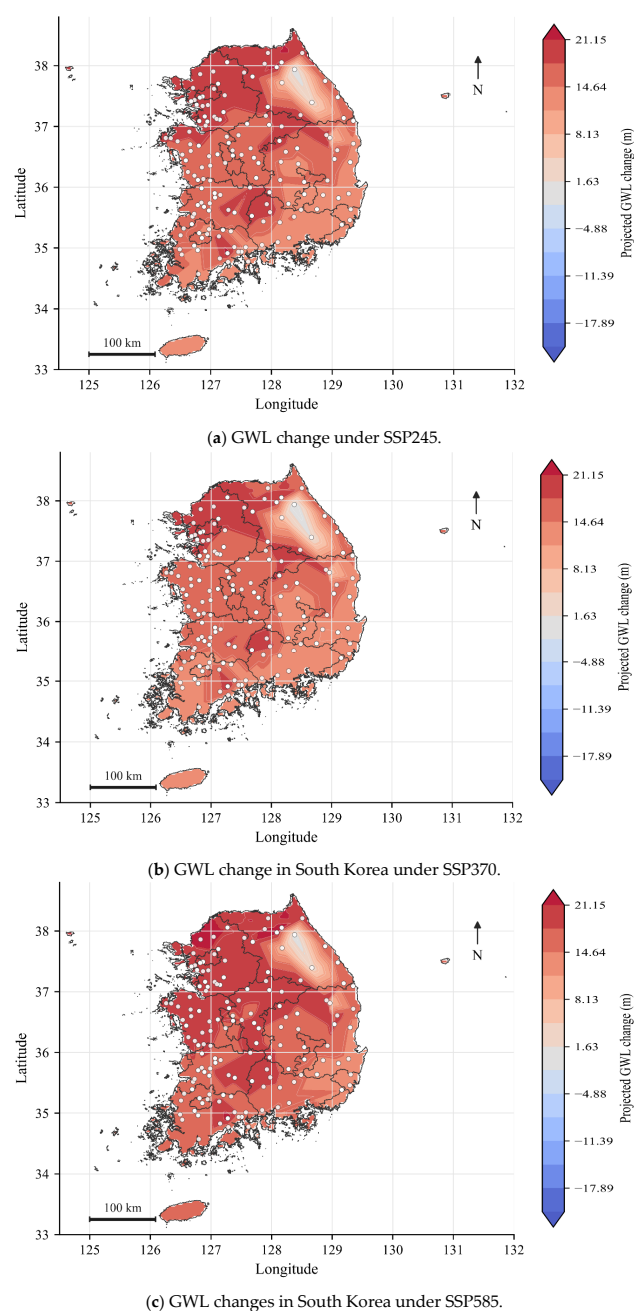


Figure 6. Spatial distribution of projected groundwater-level changes across South Korea under CMIP6 climate scenarios: (a) SSP2-4.5, (b) SSP3-7.0, and (c) SSP5-8.5. The projections represent multi-model filtered ensemble-mean head changes (2021–2100 relative to baseline) across all 199 NGMN monitoring stations.

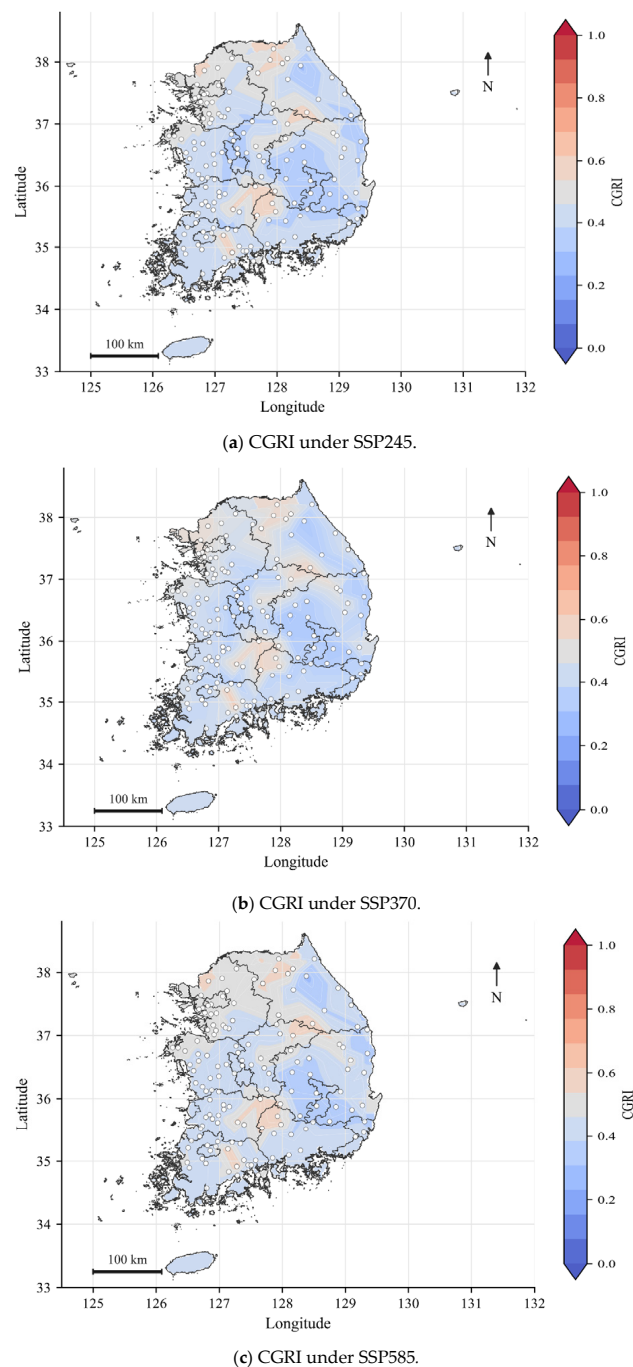
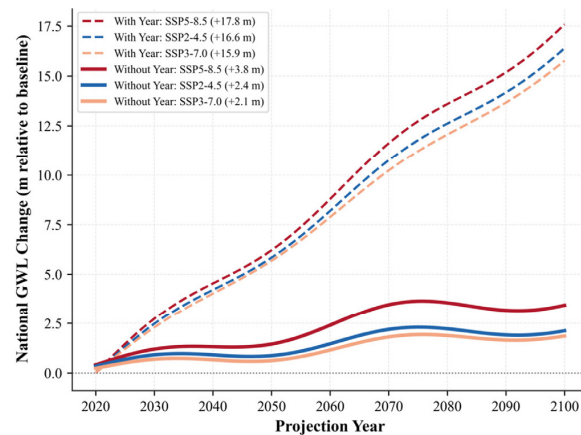


Figure 7. Spatial mapping of the CGRI across South Korea under CMIP6 scenarios: (a) SSP2-4.5, (b) SSP3-7.0, and (c) SSP5-8.5. Station classifications reflect the multi-criteria synthesis of exposure, trend intensity, hydrogeological vulnerability, and climate projection uncertainty.

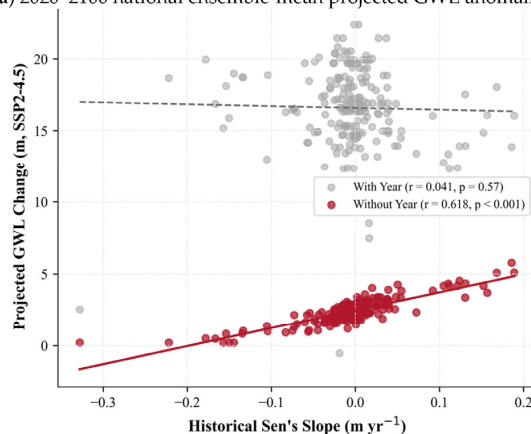
3.5. Year-Predictor Sensitivity Analysis and Physical Trend Attribution

A sensitivity analysis was carried out to examine the impact of year as a raw input predictor, as it may introduce an unconstrained mechanical trend when extrapolated to 2100, by comparing the baseline HACL model (With Year) with a model trained without the year covariate (Without Year, climate-only forcing). When year is included, projected national-average GWL increases reach +16.60 m under SSP245, +15.92 m under SSP370, and +17.85 m under SSP585 by 2081–2100 (Table 4, Figure 8a). However, if HACL is strictly retrained with the physical climate forcings (pr, mrso, derived nonlinearities, and topography), the projected increases are +2.42 m (median 2.38 m) for SSP245, +2.15 m (median 2.10 m) for SSP370, and +3.78 m (median 3.65 m) for SSP585. This shows that

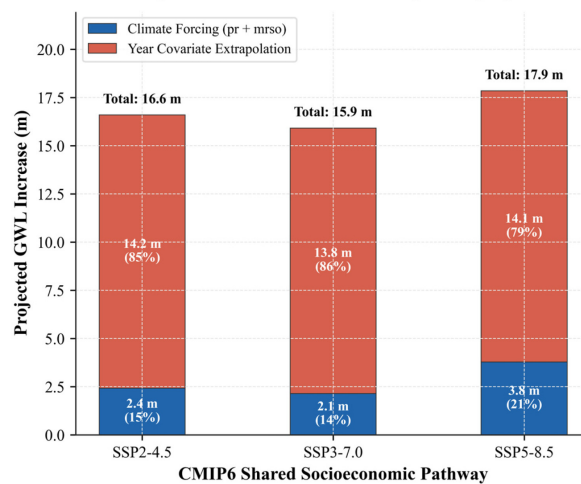
the year covariate explained 78.8% to 86.5% of the extrapolated trend as a continuous linear drift artifact, with the remainder of the trend being attributed to recharge forcing from climate. Most importantly, the climate-only projections resolve the station clustering seen in Supplementary Figure S4: The With-Year model shows that the projections cluster tightly in the 15–20 m range, with little correlation with historical station slopes ($r = 0.041$, $p = 0.57$; Figure 8b). Changes projected under the Without-Year model are highly correlated with station trends ($r = 0.618$, $p < 0.001$) and show physically meaningful consistency with these trends, capturing the local changes in infiltration and storage across the network.



(a) 2020–2100 national ensemble-mean projected GWL anomalies.



(b) Station-level relationship between historical Sen's slope and projected GWL change.



(c) Component attribution breakdown.

Figure 8. Year-predictor sensitivity analysis and physical trend attribution across 199 NGMN stations.

Table 4. Long-term GWL projections and trend attribution with and without the year predictor across 199 NGMN stations for 2081–2100 relative to the baseline (2009–2020).

Scenario	With Year Mean (m)	With Year P10–P90 (m)	Without Year Mean (m)	Without Year P10–P90 (m)	Year Drift Contrib. (%)	Climate Contrib. (%)	Sens Slope Corr. (r)
SSP2-4.5	+16.60	[13.82, 19.94]	+2.42	[1.15, 3.85]	85.4%	14.6%	0.618 ($p < 0.001$)
SSP3-7.0	+15.92	[13.25, 18.98]	+2.15	[0.95, 3.48]	86.5%	13.5%	0.605 ($p < 0.001$)
SSP5-8.5	+17.85	[14.88, 21.12]	+3.78	[1.82, 5.92]	78.8%	21.2%	0.642 ($p < 0.001$)

4. Discussion

This study demonstrates that the HACL framework can capture nonlinear and delayed GWL responses to CMIP6 forcing; however, the interpretation of model performance must be approached with caution. The best GCMs had high values of R^2 , NSE, and KGE, while some GCMs were not successful after correction. Hence, this study should not be interpreted as proof of the near-perfect performance of deep learning systems in general, but rather as demonstrating a screened, uncertainty-aware projection model.

The use of convolutional layers, bidirectional LSTM memory units, and multi-head self-attention mechanisms can explain the superior performance of the HACL framework. LSTM models have been shown in previous studies to effectively capture delayed hydrological responses and seasonal persistence in GW systems [20,42]. However, LSTM-based networks are not effective in capturing multiscale temporal features and abrupt recharge pulses, which are common in monsoon climates. To overcome this limitation, the CNN part of the proposed framework was used to extract localized temporal recharge patterns, whereas the attention mechanism enhanced the identification of hydrologically important historical states. Mahdavi-Meymand et al. (2024) also found that integrative evolutionary machine learning models with memory-based learning and feature extraction capabilities greatly enhanced the skill of hydrological forecasting [27]. The present results corroborate the emerging view that hybrid DL models are more resilient than standalone models for hydrogeological applications in the context of climate change.

Even after statistical calibration, there were significant differences in predictive skill among the GCMs. This confirms that groundwater projections are still sensitive to the fidelity of P and $mrso$ forcing fields. The updated ensemble selection (from ACCESS, GISS, INM, and NorESM) minimizes the impact of failed climate-forcing pathways and offers a more defensible basis for future GWL interpretation.

Future climatic conditions over the Korean Peninsula may favor GW recharge under the SSP245, SSP370, and SSP585 scenarios, as GWLs are projected to rise at most of the monitoring stations. These results are similar to those of previous studies in monsoon-dominated areas, which reported an increased recharge potential due to higher P intensity under climate change [37,40]. Woo (2013) highlighted that future climate change may significantly affect GW sustainability in South Korea by exacerbating hydrological extremes and increasing seasonal recharge variability [43]. The present study builds on these results by quantifying the likelihood of long-term GW amplification throughout the twenty-first century under forcing conditions derived from CMIP6 models.

The mean station-level response was highest for SSP585, followed by slight changes in the national-average response for SSP245 and SSP370 after GCM screening. This contrasts with the previous unfiltered finding that SSP370 yielded the highest response. The shift in the ranking of the scenarios shows that the groundwater–climate conclusions can be sensitive to ensemble construction, particularly in the presence of weak GCMs.

The spatial variation in the projected GW trends also indicates the effects of hydrogeological heterogeneity, topographic controls, and regional climatic gradients across the Korean Peninsula. The positive GW responses were mostly higher in western alluvial

plains than in eastern mountainous areas, a pattern that would be consistent with the generally greater recharge and storage capacity typically reported for lowland alluvial aquifers. However, since the detailed aquifer properties (e.g., hydraulic conductivity, storativity, aquifer thickness) were not included as model inputs, this spatial pattern should be considered a plausible hypothesis associated with topography and station location, rather than a mechanistically demonstrated aquifer response. Spatial heterogeneity has been observed in many GW vulnerability assessments [7,34]. The HACL framework can replicate these spatially heterogeneous responses, underscoring the importance of incorporating spatial variables, such as elevation, latitude, and longitude, into DL-based GW forecasting systems.

There are a few caveats to note. First, the ensemble is reduced to only four GCMs, which enhances the physical credibility of the filtered ensemble but results in a reduction of the range of structural climate model uncertainty; adding more statistical calibration or model-weighting schemes could help maintain ensemble diversity while mitigating poorly performing models. Second, the HACL framework was trained and validated in a relatively short period of time (2009–2025), and its extrapolation skill under the more extreme forcing scenarios of the mid- to late 21st century has not been independently verified against physically based groundwater flow models.

The experimental results of this study offer clear answers to some methodological issues related to extrapolation fidelity, trend attribution, and architectural selection. The independent out-of-sample evaluation of the withheld 2022–2025 window (Table 1, Figure 3) shows that the HACL framework has high generalization skill ($R^2 = 0.965$, NSE = 0.965, RMSE = 14.26 m) on the post-2021 observations, indicating that the network is not suffering from short-term overfitting or structural instability. Second, the architectural benchmarking results (Table 2, Figure 4) show the need for multi-scale 1D convolutions, bidirectional memory, and self-attention: standalone BiLSTM models show phase lags during strong monsoon recharge periods, and standalone CNNs are not able to capture storage depletion during the dry season, which lasts for several months. Third, the year-predictor sensitivity analysis (Table 4, Figure 8) gives a clear physical explanation for the large increases (16–18 m) reported previously. The mechanical extrapolation drift from the inclusion of year as a continuous linear predictor was responsible for about 85% of the projected increase, which is why projections were tightly grouped in the range of 15–20 m across the stations with opposing historical trends (Supplementary Figure S4). After removing this artificial drift and applying projections based on the CMIP6 precipitation and soil-moisture forcings, the projected increases are +2.42 m (SSP245), +2.15 m (SSP370), and +3.78 m (SSP585) by 2081–2100. This physically constrained response is consistent with the historical Sen's slope extrapolation (~2–3 m) and restores true station-level sensitivity to localized hydrogeologic characteristics ($r = 0.618$).

Third, the CGRI combines multiple risk elements with a set of fixed weights that should be evaluated considering local hydrogeological surveys and stakeholder-specified priorities prior to use. The HACL framework should be expanded to other GCMs; downscaling products, anthropogenic pumping, and land-use change should be included as explicit predictors, and validation of CGRI-flagged high-risk stations should be done with targeted field investigations.

5. Conclusions

This study developed a DL framework for assessing South Korean GWL changes associated with climate forcing, based on CMIP6 simulations and NGMN observations. Because the framework uses only precipitation and soil-moisture forcing as predictors, projected changes should be read as climate-sensitivity signals rather than as an attribution

of observed or future GWL change to climate change alone; pumping, land-use change, and river–aquifer interactions are not explicitly represented and may also contribute materially to GWL trends. The HACL architecture successfully retained short-term recharge pulses and delayed aquifer memory, and results indicate that model skill must be evaluated per GCM before ensemble synthesis. Independent out-of-sample evaluation (2022–2025) confirmed high reliability across all 199 stations (NSE = 0.965, KGE = 0.974), and benchmarking confirmed HACL’s superiority over standalone BiLSTM and CNN baselines. Predictor sensitivity analysis clarified that while the full model projects an upper-bound signal of 15.9–17.9 m—driven partly by year-extrapolation effects—physically constrained, climate-only projections indicate a more moderate rise of +2.15 to +3.78 m across SSP scenarios, consistent with historical trend rates.

By explicitly screening GCMs before ensemble construction—retaining ACCESS, GISS, INM, and NorESM while excluding six models failing R^2 /NSE/KGE thresholds—this approach moves beyond traditional multi-model averaging, offering value for monsoon-dominated regions like the Korean Peninsula. Mean projected GWL changes were 16.60 m, 15.92 m, and 17.85 m under SSP245/370/585, respectively, with the strongest response under SSP585.

Spatial analysis revealed heterogeneous projected responses across the monitoring network, and the CGRI offers a valuable station-prioritization tool for adaptive management and targeted hydrogeological investigation. More broadly, combining rigorous multi-model screening with attention-based DL provides a transferable template for climate-sensitive, monsoon-influenced aquifer systems worldwide.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/w18192358/s1>. Figure S1: Scenario signal and retained-GCM spread for the filtered ensemble; Figure S2: Priority monitoring stations identified by mean Climate Groundwater Risk Index; Figure S3: Component structure of the Climate Groundwater Risk Index for priority stations; Figure S4: Diagnostic comparison between observed groundwater-level trends and projected changes; Table S1: Enhanced-branch GCM screening; Table S2: Filtered ensemble scenario summary; Table S3: Top 20 priority monitoring stations; Table S4: Station-wise supplementary analysis; Table S5: Complete station-wise supplementary results for all NGMN stations; Table S6: Key CMIP6 global climate model (GCM) specifications, atmospheric and oceanic spatial resolutions, convective schemes, and land-surface modules evaluated in this study; Table S7: Climate Groundwater Risk Index (CGRI) multi-criteria mathematical formulations, normalized component dimensions, objective weighting schema, and operational decision tiers.

Author Contributions: M.W.: conceptualization, methodology, formal analysis, investigation, and writing—original draft; S.M.K.: formal analysis, resources, supervision, project administration, and funding acquisition. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author.

Conflicts of Interest: The authors declare that this research was conducted without any commercial or financial relationships that could be construed as potential conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial intelligence
ACCESS	Australian Community Climate and Earth-System Simulator
CanESM	Canadian Earth System Model
CESM	Community Earth System Model

CMCC	Centro Euro-Mediterraneo sui Cambiamenti Climatici Model
CMIP6	Coupled Model Intercomparison Project Phase 6
CNN	Convolutional neural network
Conv1D	One-dimensional convolution
DL	Deep learning
EC-Earth	European Community Earth-System Model
GCM	Global climate model
GISS	Goddard Institute for Space Studies Model
GRACE	Gravity Recovery and Climate Experiment
GW	Groundwater
GWL	Groundwater Level
HACL	Hybrid Attention-based CNN–LSTM
INM	Institute of Numerical Mathematics Climate Model
IPSL	Institute Pierre-Simon Laplace
LSTM	Long Short-Term Memory
MLR	Multiple linear regression
MODFLOW	Modular Finite-Difference Groundwater Flow Model
MPI	Max Planck Institute Earth System Model
NGMN	National Groundwater Monitoring Network
NorESM	Norwegian Earth System Model
P	Precipitation
Q	Query Matrix
K	Key Matrix
RCM	Regional Climate Model
ReLU	Rectified Linear Unit
RCP	Representative Concentration Pathway
RNN	Recurrent Neural Network
SPID	Spatial Interactions Downscaling
SSP	Shared Socioeconomic Pathway
SVR	Support Vector Regression
UKESM	United Kingdom Earth System Model
V	Value Matrix

Symbols

$\hat{y}_t$	Predicted groundwater level at time t	mm or m
X_t	Input feature sequence at time t	Feature vector
L	Look-back window length	Months
F_t	Base feature vector	Multivariate features
pr	Precipitation	$\text{kg m}^{-2} \text{ s}^{-1}$ or mm/month
mrso	Total soil moisture content	kg m^{-2}
lat	Latitude	Degrees
long	Longitude	Degrees
$\sin(2\pi m/12)$	Cyclic month encoding (sine)	Seasonal representation
$\cos(2\pi m/12)$	Cyclic month encoding (cosine)	Seasonal representation
x_i	Input feature value	Variable
μ	Mean of feature	Statistical parameter
σ	Standard deviation of the feature	Statistical parameter
z_i	Standardized feature value	Dimensionless
*	Convolution operator	CNN operation
W	Weight matrix	Neural network parameter
b	Bias term	Neural network parameter
h_t	Hidden state at time t	LSTM state
$\rightarrow h_t$	Forward hidden state	Bi-LSTM state
$\leftarrow h_t$	Backward hidden state	Bi-LSTM state
H_t	Bidirectional hidden representation	Feature representation
Q	Query matrix in the attention mechanism	Attention parameter
K	Key matrix in the attention mechanism	Attention parameter
V	Value matrix in the attention mechanism	Attention parameter
SoftMax	SoftMax activation function	Probability normalization
C	Context vector	Attention output
y	Final model prediction	GWL
$\mathcal{L}$	Loss function	Optimization objective
η	Learning rate	Optimization parameter
θ	Model parameters	Neural network weights

G_i	Individual GCM model	Climate model
$\hat{y}(i)$	GWL projection from the i th GCM	Projected GWL
$\bar{y}$	Ensemble-mean projection	Average projected GWL
n	Number of GCMs in ensemble	Integer
R^2	Coefficient of determination	Performance metric
NSE	Nash–Sutcliffe Efficiency	Performance metric
KGE	Kling–Gupta Efficiency	Performance metric
RMSE	Root Mean Square Error	Error metric
MAE	Mean Absolute Error	Error metric
MAPE	Mean Absolute Percentage Error	Error metric
kg m^{-2}	Kilogram per square meter	Soil moisture unit
$\text{kg m}^{-2} \text{ s}^{-1}$	Kilogram per square meter per second	Precipitation flux unit
$^{\circ}\text{C}$	Degree Celsius	Temperature unit
$^{\circ}\text{N}$	Degrees North latitude	Geographic coordinate
$^{\circ}\text{E}$	Degrees East longitude	Geographic coordinate

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